

Introduction:

1. Source of the Dataset:

The dataset used is the UCI Heart Disease dataset from the UCI Machine Learning Repository. It contains clinical attributes such as age, cholesterol, blood pressure, chest pain type, and heart rate, along with a target variable indicating the presence or absence of heart disease.

2. Application Domain:

This dataset belongs to the healthcare domain and is used for cardiovascular disease analysis, risk assessment, and prediction. It helps in understanding medical factors that contribute to heart disease and supports clinical research.

3. Knowledge that can be Extracted:

The dataset helps identify relationships between clinical features and heart disease. It reveals important risk factors, patterns, and correlations that can be used to understand disease occurrence and patient risk levels.

4. Real-World Decisions Supported:

The insights can support early diagnosis, risk prediction, preventive care planning, and clinical decision-making, helping healthcare professionals improve patient treatment and outcomes.

Dataset Description:

1. Definition

Data Object:

A data object represents a single patient record in the dataset. Each object contains medical and clinical measurements related to heart health, along with a target variable indicating whether the patient has heart disease or not.

Attributes/Features:

Attributes are the variables that describe each patient. These include demographic features (age, sex), clinical measurements (cholesterol, resting blood pressure, maximum heart rate), and diagnostic indicators (chest pain type, exercise-induced angina, ST depression, etc.). The target attribute represents the presence (1) or absence (0) of heart disease.

2. Dataset Representation

The dataset can be represented as a feature vector:

$$X = (x_1, x_2, x_3, \dots, x_d)$$

Where:

- X represents a patient record
- x_1 = age
- x_2 = sex
- x_3 = chest pain type (cp)
- x_4 = resting blood pressure (trestbps)
- x_5 = cholesterol (chol)
- x_6 = fasting blood sugar (fbs)
- x_7 = resting ECG (restecg)
- x_8 = maximum heart rate (thalach)
- x_9 = exercise-induced angina (exang)
- x_{10} = ST depression (oldpeak)
- x_{11} = slope of ST segment (slope)
- x_{12} = number of major vessels (ca)
- x_{13} = thalassemia (thal)

Target variable:

$$y = \text{target} \in \{0,1\}$$

3. Attribute Classification Table

Attribute	Type	Category	Example Values
age	Numerical	Ratio	29, 45, 63
sex	Binary	Nominal	0 = Female, 1 = Male
cp (chest pain type)	Nominal	Categorical	0, 1, 2, 3
trestbps	Numerical	Ratio	120, 140, 160
chol	Numerical	Ratio	200, 240, 300
fbs	Binary	Nominal	0 = False, 1 = True
restecg	Nominal	Categorical	0, 1, 2

Attribute	Type	Category	Example Values
thalach	Numerical	Ratio	120, 150, 180
exang	Binary	Nominal	0 = No, 1 = Yes
oldpeak	Numerical	Ratio	0.0, 1.5, 3.2
slope	Ordinal	Categorical	0, 1, 2
ca	Numerical	Discrete Ratio	0, 1, 2, 3
thal	Nominal	Categorical	0, 1, 2, 3
target	Binary	Class label	0 = No disease, 1 = Disease

Attribute Type Classification Summary

Nominal Attributes:

cp, restecg, thal

Ordinal Attributes:

slope

Numerical Attributes (Interval/Ratio):

age, trestbps, chol, thalach, oldpeak, ca

Binary Attributes:

sex, fbs, exang, target

Data Objects & Attributes Analysis:

Important Attributes:

Based on the correlation analysis, distribution plots, and similarity analysis, several attributes appear highly informative for predicting heart disease. These include chest pain type (cp), maximum heart rate achieved (thalach), ST depression (oldpeak), number of major vessels (ca), thalassemia (thal), and exercise-induced angina (exang). These features showed strong correlation with the target variable and clear separation between patients with and without heart disease. Age and resting blood pressure (trestbps) also showed moderate influence.

Redundant Features:

Some attributes appear less informative and may be considered partially redundant. Fasting

blood sugar (fbs) and resting ECG (restecg) showed weak correlation with the target variable and limited contribution to distinguishing between classes. Cholesterol (chol) showed moderate variation but weaker predictive power compared to other features.

Nature of Relationships:

Both linear and nonlinear relationships were observed between the attributes and the target variable. Features such as thalach, oldpeak, and ca showed relatively linear relationships with heart disease risk. However, categorical attributes such as cp, thal, and exang showed nonlinear relationships, where specific categories were strongly associated with disease presence. This indicates that the dataset contains a combination of linear and nonlinear patterns.

Similarity Analysis:

Pearson and Spearman Correlation

Pearson correlation was used to measure the linear relationship between the attributes and the target variable, while Spearman correlation was used to measure monotonic relationships based on rank. The results showed that attributes such as chest pain type (cp), maximum heart rate (thalach), ST depression (oldpeak), and number of major vessels (ca) had strong correlation with the target variable. Pearson correlation indicated clear linear relationships for several numerical attributes, especially thalach and oldpeak.

Spearman correlation values were slightly higher for some attributes, indicating the presence of nonlinear but monotonic relationships. This suggests that certain features influence heart disease risk in a nonlinear manner. Overall, both correlation measures confirmed that cp, thalach, oldpeak, ca, and thal are among the most informative attributes.

Euclidean Distance

Euclidean distance was computed for a subset of 10 patients to measure the dissimilarity between their medical profiles. Patients with smaller Euclidean distance values were more similar in terms of clinical attributes, while larger distance values indicated greater differences in health conditions. The dissimilarity matrix showed that some patients formed clusters with similar feature values, suggesting similar heart disease risk profiles.

Cosine Similarity

Cosine similarity was used to measure similarity based on the orientation of feature vectors rather than magnitude. Higher cosine similarity values indicated that patients had similar overall patterns in their medical attributes. This measure was useful in identifying patients with similar health profiles even when their absolute values differed.

Jaccard Similarity (Binary Attributes)

Jaccard similarity was calculated using binary attributes such as sex, fasting blood sugar (fbs), exercise-induced angina (exang), and target. The results showed varying levels of similarity between patients based on shared medical conditions. Higher similarity values indicated patients with similar binary clinical characteristics, while lower values indicated differences in medical conditions.

Dissimilarity Matrix Interpretation

The dissimilarity matrix created using Euclidean distance showed clear variation in similarity among patients. Some patients had low dissimilarity values, indicating similar clinical profiles and potentially similar disease risk. Others showed high dissimilarity, reflecting differences in medical attributes. These results demonstrate that similarity measures can help identify patient groups with similar health characteristics and support clustering, classification, and risk prediction in healthcare applications.

Data Quality Study

Missing Values

The dataset was examined for missing values using statistical checks and visualization techniques. The analysis showed that there were no significant missing values present in the dataset, and most attributes contained complete records. The missing data visualization confirmed that the dataset is largely complete, ensuring reliable analysis and model training without the need for extensive missing value treatment.

Noise and Outliers

Outlier detection was performed using boxplots and statistical measures. Some attributes, particularly cholesterol (chol), resting blood pressure (resttbps), and ST depression (oldpeak), showed the presence of outliers. These outliers represent extreme medical values, which may correspond to patients with severe or unusual health conditions. While these values are valid in a medical context, they may influence statistical analysis and model performance.

Inconsistent Entries

The dataset was checked for inconsistent and invalid values by examining attribute ranges and unique values. All categorical and binary attributes, such as sex, chest pain type (cp), fasting blood sugar (fbs), and target, contained valid and expected values within their defined ranges. No major inconsistencies or invalid entries were observed, indicating good overall data consistency.

Irrelevant Attributes

Correlation analysis and feature evaluation showed that some attributes, such as fasting blood sugar (fbs) and resting ECG (restecg), had relatively weak relationships with the target variable. These attributes may have limited predictive importance compared to other features such as chest

pain type, maximum heart rate, and ST depression. However, they may still contribute useful information in combination with other features.

Missing Data Visualization

Missing data distribution was visualized using heatmaps and bar plots, which confirmed that the dataset contains minimal or no missing values. These visualizations provided a clear overview of data completeness and verified that the dataset is suitable for further preprocessing and analysis.

Preprocessing Methods:

Missing Value Imputation

To handle missing values and ensure data completeness, three imputation methods were applied:

1. Mean/Median Imputation:

Mean and median imputation methods were used to replace missing values with the average or median of the respective attribute. Median imputation was particularly useful for attributes containing outliers, as it is less sensitive to extreme values. This method ensured that missing values were filled using statistically representative values from the dataset.

2. KNN Imputation:

KNN (K-Nearest Neighbors) imputation was applied to replace missing values based on the similarity between patient records. This method identifies the nearest neighbors using distance measures and fills missing values using information from similar data objects. It preserves the underlying structure and relationships in the dataset more effectively than simple statistical imputation.

3. Iterative Imputation:

Iterative imputation was used as an advanced method that predicts missing values using regression-based techniques. Each attribute with missing values was treated as a dependent variable and estimated using other attributes as predictors. This method provided more accurate and realistic imputed values by considering relationships among multiple features.

Preprocessing Pipeline

A preprocessing pipeline was created using SimpleImputer, StandardScaler, and LogisticRegression to automate the preprocessing and model training process.

SimpleImputer:

This step replaces missing values using a statistical strategy such as median imputation, ensuring complete data for model training.

StandardScaler:

Scaling was applied to normalize the feature values so that all attributes contribute equally to the

model. This prevents attributes with larger numerical ranges from dominating the learning process.

LogisticRegression:

The Logistic Regression model was used as the classification algorithm to predict the presence or absence of heart disease based on the processed features.

Importance of Pipelines

Pipelines are important because they automate the entire preprocessing and modeling workflow. They ensure that all preprocessing steps are applied consistently and in the correct order. Pipelines also make the process more efficient, organized, and reproducible. Additionally, they simplify model deployment by combining multiple steps into a single unified structure.

Prevention of Data Leakage

Pipelines help prevent data leakage by ensuring that preprocessing steps such as imputation and scaling are performed only on the training data and then applied to the test data. This prevents information from the test set from influencing the training process. As a result, the model evaluation remains unbiased and reflects the true performance of the model on unseen data.

Results and Discussion:

1. Patterns Discovered

The analysis revealed several important patterns related to heart disease risk. Patients with higher ST depression (oldpeak), exercise-induced angina (exang), and a greater number of major vessels (ca) were more likely to have heart disease. Additionally, patients with lower maximum heart rate (thalach) showed higher disease risk. Chest pain type (cp) also showed a strong relationship with the target variable, indicating that certain chest pain categories are more associated with heart disease.

2. Most Informative Attributes

Based on correlation analysis, distribution plots, and similarity measures, the most informative attributes were chest pain type (cp), maximum heart rate (thalach), ST depression (oldpeak), number of major vessels (ca), thalassemia (thal), and exercise-induced angina (exang). These attributes showed strong relationships with the target variable and played a significant role in distinguishing between patients with and without heart disease. Attributes such as fasting blood sugar (fbs) and resting ECG (restecg) were found to be less informative.

3. Effect of Preprocessing on Insights

Preprocessing improved the overall quality and reliability of the analysis. Missing value imputation ensured that no incomplete records affected the results. Scaling standardized the feature values, allowing fair comparison between attributes. The preprocessing pipeline ensured

consistent data transformation and prevented data leakage, resulting in more accurate and reliable model performance and interpretation.

4. Healthcare Decisions Supported

The discovered patterns can support several healthcare decisions. Medical professionals can use these insights for early detection of heart disease risk and identify high-risk patients based on key clinical indicators. This can help in recommending preventive measures, conducting further diagnostic tests, and planning appropriate treatment. Additionally, predictive models developed using this data can assist in clinical decision support systems, improving diagnosis accuracy and patient care.

Conclusion:

This study analyzed the UCI Heart Disease dataset using data mining and preprocessing techniques to discover meaningful patterns and relationships between clinical attributes and heart disease risk. The analysis identified key attributes such as chest pain type, maximum heart rate, ST depression, number of major vessels, and exercise-induced angina as the most informative predictors of heart disease. Correlation and similarity analysis confirmed the presence of both linear and nonlinear relationships among the features.

Data quality assessment showed that the dataset was largely complete, with some outliers present in certain medical attributes. Missing value imputation and preprocessing methods improved data consistency and reliability. The preprocessing pipeline ensured proper handling of missing values, feature scaling, and model training while preventing data leakage.

The results demonstrate that data mining techniques can effectively identify important risk factors and support predictive modeling in healthcare. These insights can assist medical professionals in early diagnosis, risk assessment, and clinical decision-making, ultimately contributing to improved patient care and better healthcare outcomes.